VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI– 590018, KARNATAKA, INDIA

REPORT

ON

"PHYSICAL EDUCATION (SPORTS AND ATHLETICS)-I"

Submitted in partial fulfillment of the requirements for the award of degree of BACHELOR OF ENGINEERING IN

COMPUTER SCIENCE AND ENGINEERING-DATA SCIENCE

Submitted By

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CERTIFICATE

Certified that Physical Education (Sports and Athletics)-I Report, work entitled "**On Subject of 6th Sem, Physical Education (Sports and Athletics)-I"** carried out by RUCHITHA R (1KN22CD045) a bonified student of KNS Institute of Technology, Bengaluru, during the year 2024-25. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Physical Education (Sports and Athletics)-I report deposited in the departmental library.

The Physical Education (Sports and Athletics)-I report has been approved as it satisfies the academic requirements in respect of Physical Education prescribed for the said Degree.

UNDER THE GUIDANCE OF

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ABSTRACT

The purpose of this study was to access the physical fitness of students and improve them. Physical education is a process that uses physical activity as a means to help people acquire skills, fitness, knowledge, and attitudes that contribute to their optimal development and well-being.

Students were given a brief introduction on healthy lifestyle, physical fitness and wellness. Proper food and nutrition play a crucial role in physical and mental development of the body. A pre-fitness test was conducted to evaluate students' physical fitness and identify the areas that need improvement.

We learned about the different warm up exercises and the importance of warm up exercises before playing any sport. Demonstration of push ups and pull ups were given. Students were given different types of tests like 30 m dash, shuttle run, etc. We were taught about the different postural deformities that occur due to lack of physical movements and training. Stress management is important to maintain a healthy mind and body. Many Athletics games like Running, Shotput, Disc, Relay, etc. were played during the course of physical training.

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1.INTRODUCTION

PHYSICAL EDUCATION

DIFINATION

Physical Education (PE) is the process of developing and maintaining physical fitness, health, and well-being through sports, games, and other physical activities. It plays a vital role in promoting overall health and fitness among individuals. Physical Education (PE) is an essential component of a well-rounded education, focusing on the development of physical fitness, motor skills, and overall well-being. It promotes physical activity, encourages healthy lifestyles, and helps students learn the importance of teamwork, discipline, and perseverance. PE programs aim to improve physical health, enhance mental well-being, and foster social skills by providing students with the opportunity to engage in various sports and fitness activities. Additionally, PE contributes to developing lifelong habits of exercise and healthy living, which can have a lasting positive impact on individuals' lives.

OBJECTIVE

The primary objectives of Physical Education are to develop physical fitness and coordination, improve health and well-being, enhance teamwork and communication skills, foster sportsmanship and fair play, and develop emotional and social skills. By achieving these objectives, individuals can lead healthier and more productive lives.

Physical Education encompasses various components, including sports, athletics, games, and fitness. Sports and athletics provide opportunities for individuals to develop teamwork and competitive skills, while games and fitness activities promote recreational and lifelong physical activity.

Through Physical Education, individuals can experience numerous benefits, including Dand emotional skills, fostered teamwork and communication, and preparation for lifelong physical activity and health.

MODULE-4

ORIENTATION

1.1 POSTURAL DEFORMITIES

Postural deformities are problems in the normal alignment of the body. When a person's posture is not correct for a long time, it can lead to changes in the shape of the spine or other body parts. This can cause pain, tiredness, and difficulty in doing physical activities.Postural deformities are problems in body posture where bones or muscles are not in theright position. This can lead to pain, discomfort, and difficulty in movement.

FIG:1.1 POSTURAL DEFORMITIES

1.1.1 Importance of Postural Deformities

1. **Early Detection and Prevention**
   - Recognizing postural problems early helps prevent serious health issues in the future.
   - Children and teenagers are still growing, so early correction can stop deformities from becoming permanent.
2. **Improves Physical Performance**
   - Good posture improves balance, strength, flexibility, and body coordination.
   - Athletes can perform better in sports and avoid injuries by maintaining proper posture.
3. **Reduces Pain and Discomfort**
   - Poor posture often leads to back pain, neck pain, and muscle stiffness.
   - Understanding postural deformities helps in choosing the right exercises or treatments to reduce pain.
4. **Boosts Confidence and Appearance**
   - Correct posture improves body shape and makes a person look taller and more confident.Many people feel better about themselves when they stand or sit properly.

5. **Supports Daily Activities**
   - Normal tasks like walking, sitting, lifting, or playing sports become easier and safer with good posture.
   - People with correct posture get tired less quickly and are more active throughout the day.
6. **Helpful in Education and Career**
   - Students who sit for long hours need to know the importance of posture to avoid long-term problems.
   - In careers like teaching, sports, nursing, or driving, maintaining posture is part of professional health and safety.
7. **Promotes Healthy Lifestyle**
   - Learning about posture encourages people to stay active, stretch, and exercise regularly.
   - It spreads awareness about fitness, body care, and injury prevention.

1.1.2 CAUSES OF POSTURAL DEFORMITIES

- Poor Postural Habits
- Lack of Physical Activity
- Weak Muscles
- Carrying Heavy Bags
- Improper Furniture
- Injury or Accidents
- Genetic or Birth Defects
- Nutritional Deficiency
- Obesity

1.1.3 HOW TO MAINTAIN THE GOOD POSTURE

To maintain good posture and prevent postural deformities, it is important to develop healthy habits in daily life. One should sit and stand properly by keeping the back straight, shoulders relaxed, and feet flat on the ground. Carrying heavy school bags on one shoulder should be avoided, and a backpack with two straps should be used instead. Regular physical activity, such as stretching, yoga, and walking, helps to strengthen the muscles that support the spine. Using proper furniture that supports the body and maintaining the correct position while using a phone or computer is also important. Taking breaks while studying or sitting for long hours, sleeping on a firm mattress, and staying active throughout the day can all help in improving posture. If any pain or discomfort is noticed, medical advice should be taken to avoid long-term problems.

1.2. STRESS MANAGEMENT:

Stress management involves the use of techniques and lifestyle changes to reduce and cope with physical and emotional tension caused by stress. It is essential for maintaining both mental and physical health, especially in today's fast-paced world. Common methods include physical activities like regular exercise and yoga, which help release endorphins and reduce stress hormones. Relaxation techniques such as deep breathing, meditation, and progressive muscle relaxation calm the mind and body. Managing time effectively, setting priorities, and maintaining a balanced routine also help prevent feeling over whelmed. Building a strong support system through family, friends, or professional counsellors provides emotional relief and perspective. Additionally, healthy habits like getting enough sleep, eating a balanced diet, and limiting caffeine or alcohol intake contribute to better stress control. By adopting these strategies, individuals can improve their ability to handle challenges, enhance overall well being, and prevent the negative effects of chronic stress.

FIG:1.2 STRESS MANAGEMENT

1.2.1 Stress Management Techniques

Managing stress effectively involves a combination of physical, mental, and emotional strategies that help the body and mind relax and recover from daily pressures. One of the most effective techniques is physical activity, such as walking, running, yoga, or swimming. Regular exercise helps reduce levels of the stress hormone cortisol and increases the production of endorphins, which are natural mood lifters. Even light physical activity can lead to significant improvements in mood and energy levels.

Another powerful method is relaxation and mindfulness. Techniques like deep breathing, progressive muscle relaxation, and meditation allow the mind to slow down and break the cycle of constant stress. Mindfulness, in particular, encourages individuals to focus on the

present moment without judgment, helping reduce anxiety and improve emotional control. Practices such as yoga and tai chi combine movement with mindfulness and are especially effective in calming both the mind and body.

FIG:1.3 STRESS MANAGEMENT TECHNIQUES

- MCognitive and emotional techniques also play a key role in managing stress. These include positive thinking, reframing negative thoughts, and practicing gratitude. Time management is equally important—learning to prioritize tasks, avoid procrastination, and set realistic goals can reduce feelings of overwhelm.
- Seeking social support—talking to friends, family, or professionals—can also lighten emotional burdens and provide helpful perspectives. Incorporating healthy lifestyle habits like getting enough sleep, maintaining a balanced diet, staying hydrated, and limiting stimulants such as caffeine or alcohol can further improve resilience to stress.
- Over time, building a routine that includes these techniques not only reduces current stress levels but also enhances long-term mental and physical well-being.
- stress is an unavoidable part of life, but with proper management techniques, its impact can be significantly reduced. By adopting practices such as regular physical activity, mindfulness, time management, and maintaining a strong social support system, individuals can build resilience and maintain mental well-being.
- Understanding stress and learning to manage it effectively is essential for a healthy and productive life.

MODULE-V
TABLE TENNIS

2.1 INTRODUCTION

- Table Tennis, also known as ping-pong, is a fast-paced indoor racket sport played with a lightweight ball and small paddles.The game involves two or four players hitting the ball across a table divided by a net, aiming to score points by outplaying the opponent.
- The sport began in Victorian England during the 1880s as a miniature version of lawn tennis, played indoors.Initially, players used books as nets and household items like cigar box lids as paddles.
- The International Table Tennis Federation (ITTF) was established in 1926 to create global rules and structure.That same year, the first World Championships were held in London.
- Table Tennis became an Olympic sport in 1988 during the Seoul Games.It has since become hugely popular in Asia, with China, Japan, and South Korea leading in international performance.
- The sport is loved for its quick reflexes, spin techniques, and tactical gameplay.It is accessible to all age groups, requires minimal space, and promotes both physical and mental fitness.
- Table Tennis is widely played in schools, colleges, clubs, and homes as a recreational activity.It helps improve hand-eye coordination, boosts concentration, and encourages social interaction.

FIG:2.1

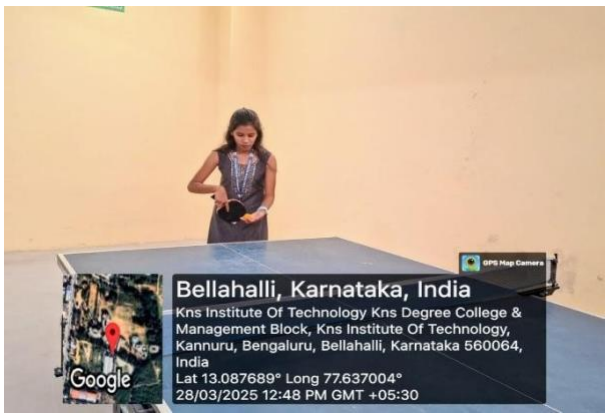

FIG:2.2

2.2 RULES AND SCORING SYSTEM

- **Size and Weight:** A standard tennis ball must have a diameter between **6.54 cm and 6.86 cm**, and it should weigh between **56.0 grams and 59.4 grams**. This ensures uniformity and fair play across all levels of the game.
- **Bounce (Rebound Test):** When dropped from a height of **254 cm (100 inches)** onto a concrete surface, the tennis ball must rebound to a height between **135 cm and 147 cm**.

This bounce test ensures that the ball has the correct internal pressure and responsiveness.

- **Deformation:** The ball is tested for deformation under pressure. The **forward deformation** must be between **0.56 cm and 0.74 cm**, and the **return deformation** should range from **0.89 cm to 1.08 cm**. These values ensure that the ball maintains its shape and performance during play.
- **Types of Balls:** According to the ITF, there are three types of tennis balls. **Type 1 (Fast)** is used for slower courts like clay, **Type 2 (Medium)** is standard and used on most courts, and **Type 3 (Slow)** is for faster courts such as grass or artificial turf.
- **Color:** The ball must be either **white or yellow**, although **optic yellow** is the standard color used in most professional matches for better visibility.
- **Pressure:** Tennis balls are usually **pressurized** to provide the correct bounce and speed. However, **pressureless balls** are also used, mainly for training purposes, as they are more durable and last longer.
- **Number of Balls in Match:** In official matches, either **three or six balls** are used. Balls are changed after every **7 games initially**, and then every **9 games** to maintain performance and consistency.
- **ITF Approval:** Only tennis balls approved by the **International Tennis Federation (ITF)** can be used in official tournaments. Organizers must choose balls from the ITF's list of certified balls.
- **Condition and Storage:** Tennis balls must be clean and in good condition before play. They are usually stored in **pressurized cans** to maintain their bounce and durability until use.
- **Identification:** Balls are often marked with **numbers or logos** to distinguish between different sets during a match, helping both players and officials track ball usage.

2.3 EQUIPTMENTS

FIG:2.3

FIG:2.4

- ✓ According to the rules set by the **International Table Tennis Federation (ITTF)**, the standard size of the table is **2.74 meters (9 feet) in length**, **1.525 meters (5 feet) in width**, and **76 centimeters (2.5 feet) in height** from the ground.
- ✓ The table is divided into two equal halves by a **net** that is **15.25 centimeters (6 inches)** high. For doubles matches, a white center line divides each side into two equal parts to indicate the service areas.
- ✓ The surface of the table is usually dark-colored (blue or green) with a **matte finish** to reduce glare and improve visibility of the ball. The edges of the table are marked with **white lines**: a 2 cm wide line along all four edges and a 3 mm wide center line for doubles.
- ✓ The material of the table should allow the ball to bounce approximately **23 cm** when dropped from a height of **30 cm**.
- ✓ A good table tennis table must be placed on a level, stable surface and in an area with sufficient space around it for movement.
- ✓ Proper lighting and surroundings also contribute to better play. Whether in professional tournaments or casual play, the quality and condition of the table significantly affect the speed, spin, and control of the game.

2.4 TECHNIQUES AND PLAYING STYLES

Players use various grips like the shakehand and penhold. Common strokes include:

1.FOREHAND/BACKHAND DRIVE:

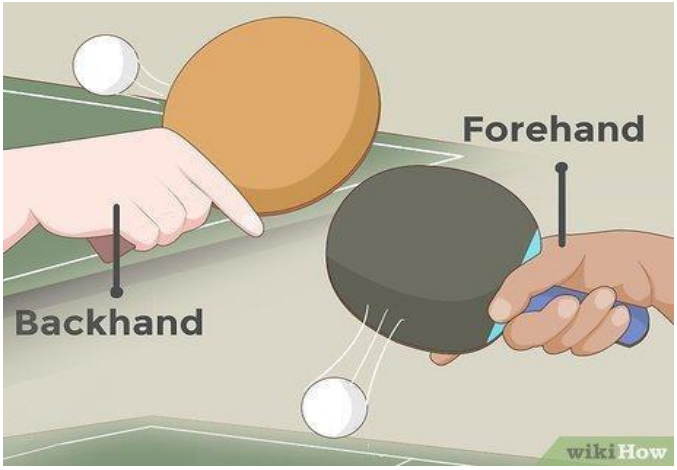

FIG:2.5 FOREHAND/BACKHAND

- The forehand drive is executed by swinging the racket across the body with the palm facing forward, generating speed and topspin. It is used when the ball comes to the dominant side of the player and allows for powerful, aggressive returns.
- The backhand drive involves hitting the ball on the non-dominant side, using a shorter and more compact stroke with the back of the hand facing the direction of the ball.

2. LOOP(SPIN-HEAVY STROKE):

- The loop is an offensive stroke that applies heavy topspin to the ball, making it dip quickly after crossing the net.
- It is commonly used against backspin balls to lift and attack them effectively.
- It can be executed on both the forehand and backhand sides, requiring good timing and body rotation.

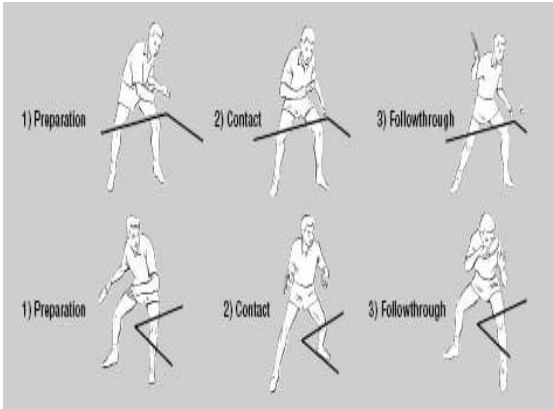

FIG:2.6 LOOP

3.CHOP (DEFENSIVE UNDERSPIN):

- The chop is a defensive stroke used to return fast or spinny shots with heavy underspin, causing the ball to slow down and drop.
- It is often played away from the table to control the pace and disrupt the opponent's rhythm.
- The racket moves in a downward slicing motion, brushing underneath the ball to generate spin.
- Proper wrist action and timing are crucial for controlling the spin and depth of the shot.

FIG:2.7 CHOP

4.SMASH

A smash is a high-speed, forceful shot used to finish points, typically executed when the ball is high or weak.It aims to overwhelm the opponent with speed, making it difficult to return.The player uses a full arm swing, often with a forehand motion, to hit the ball with maximum power.Good timing, footwork, and balance are essential to ensure accuracy and control.

FIG:2.8 SMASH

2.5 COMPETITIONS AND GLOBAL INFLUENCE OF THE TABLE TENNIS

- Olympic Games: Table Tennis became an Olympic sport in 1988 at the Seoul Games. Events include men's and women's singles and team competitions, with mixed doubles added in 2020.
- World Table Tennis Championships: Organized by the International Table Tennis Federation (ITTF) since 1926, this is one of the oldest and most respected competitions in the sport.
- ITTF World Cup and World Tour: These tournaments feature the top-ranked players globally and provide valuable ranking points. The WTT (World Table Tennis) series, launched in recent years, aims to modernize and promote professional table tennis worldwide.
- Continental and Regional Championships: Events like the Asian Table Tennis Championships, European Championships, and Commonwealth Games play vital roles in developing regional talent.

AEROBICS

3.1 INTRODUCTION:

- Aerobics is a form of physical exercise that combines rhythmic aerobic exercise with stretching and strength training routines with the goal of improving all elements of fitness (flexibility, muscular strength, and cardio-vascular fitness).
- It is usually performed to music and may be practiced in a group setting led by an instructor (fitness professional). With the goal of preventing illness and promoting physical fitness, practitioners perform various routines.
- Formal aerobics classes are divided into different levels of intensity and complexity and will have five components: warm-up (5–10 minutes), cardiovascular conditioning (25–30 minutes), muscular strength and conditioning (10–15 minutes), cool-down (5–8 minutes) and stretching and flexibility (5–8 minutes).
- Aerobics classes may allow participants to select their level of participation according to their fitness level. Many gyms offer different types of aerobic classes. Each class is designed for a certain level of experience and taught by a certified instructor with a specialty area related to their particular class.

FIG:3.1 AEROBICS

3.1.2MOVES AND TECHNIQUES:

- ften moves are referred to as Reebok step moves.
- The "basic" step involves raising one foot onto the step, then the other so that they are both on the step, then stepping the first foot back, followed by the second. A "right basic" would involve stepping right foot up, then the left, then returning to the floor alternating right then left.

- Some instructors switch immediately between different moves, for example between a right basic and a left basic without any intervening moves, effectively "tapping" the foot without shifting weight; tap-free or smooth stepping alternates the feet without "taps"

- Regular participation in aerobics offers numerous health benefits. It improves cardiovascular endurance, helps in weight management, strengthens muscles, boosts mood, and enhances overall energy levels. Additionally, it promotes better coordination, flexibility, and balance. Aerobics is often done in group classes, making it a fun and social activity, though it can also be done individually at home or in a gym setting

FIG:3.2

- Improves Cardiovascular Health: Aerobics strengthens the heart and lungs, enhancing blood circulation and oxygen delivery throughout the body.
- Aids Weight Management: Regular aerobic exercise helps burn calories and reduces body fat, supporting weight loss and maintenance.
- Increases Stamina and Endurance: With consistent practice, aerobic exercise boosts energy levels and reduces fatigue during daily activities.
- Enhances Mood and Mental Health: Aerobic activity stimulates the release of endorphins, which help reduce stress, anxiety, and depression.
- Controls Blood Sugar Levels: Aerobic workouts can help regulate insulin and lower blood sugar, which is beneficial for managing diabetes.
- Improves Lung Capacity: Regular aerobic activity enhances lung function and respiratory efficiency.
- Boosts Immune System: It increases circulation of immune cells, helping the body detect and fight illnesses more effectively.
- Promotes Better Sleep: People who engage in regular aerobic activity often fall asleep faster and enjoy deeper rest.
- Supports Joint Health: Low-impact aerobic exercises like swimming and walking help maintain joint flexibility and reduce stiffness.

CONCLUSION:

- Physical Education (Sports & Athletics)
- plays a pivotal role in promoting overall health, fitness,andwell-being.Physical education fosters physical,mental,and social development. It cultivates essential life skills such as teamwork, discipline, and resilience.
- Importance of Sports and Athletics: Sports enhance cardiovascular fitness, strength, and flexibility. They provide an outlet for stress relief and promote a positive mindset. Athletics encourage fair play, sportsmanship, and healthy competition.
- Inclusivity and Diversity: Physical education should be accessible to all, regardless of age, gender, or ability. Celebrate diversity by embracing various sports and activities.

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